

Paper 1 - Set 07 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Indices and surds
2	D	Statistics
3	D	Functions and transformations
4	C	Quadratic functions
5	F	Probability
6	D	Trigonometric functions and equations
7	E	Indices and surds
8	B	Triangles, radians and sectors
9	C	Probability
10	E	Quadratic functions
11	A	Functions and transformations
12	D	Statistics
13	D	Differentiation
14	A	Sequences and series
15	D	Graphs of functions
16	E	Simultaneous equations and inequalities
17	F	Graphs of functions
18	E	Sequences and series
19	B	Simultaneous equations and inequalities
20	E	Integration

Worked solutions

Question 1 IND-02 · Indices and surds · spec MM1.1

Find the value of $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$.

Answer: C

The negative index inverts the fraction: $\left(\frac{8}{27}\right)^{2/3}$. Cube root first: $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$.

Question 2 STA-08 · Statistics · spec M6.3

A cumulative frequency graph for 80 observations passes through (20, 20), (30, 40) and (45, 60). Estimate the interquartile range.

Answer: D

Lower quartile at cumulative frequency 20: value 20. Upper quartile at 60: value 45. IQR = 25.

Question 3 FUN-14 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ has a maximum point at (3, 8). The graph of $y = f(x - a) + b$ has its maximum at (1, 2). Find $a + b$.

Answer: D

The maximum moves from (3, 8) to $(3 + a, 8 + b) = (1, 2)$, so $a = -2$, $b = -6$ and $a + b = -8$.

Question 4 QUA-15 · Quadratic functions · spec MM1.3

The graph of $y = x^2 + bx + c$ crosses the x -axis at (1, 0) and (5, 0). What is the y -coordinate of the vertex?

Answer: C

$y = (x - 1)(x - 5)$, vertex at $x = 3$: $y = (2)(-2) = -4$.

Question 5 PRO-07 · Probability · spec M7.6

Three fair coins are tossed. Find the probability of getting more heads than tails.

Answer: F

Need 2 or 3 heads: $\frac{3+1}{8} = \frac{1}{2}$ (by symmetry between heads and tails).

Question 6 TRG-03 · Trigonometric functions and equations · spec MM4.3

Find the exact value of $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$.

Answer: D

$\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = 1$.

Question 7 IND-17 · Indices and surds · spec MM1.1

Solve $x^{\frac{3}{2}} = 8x^{-\frac{3}{2}}$ for $x > 0$.

Answer: E

Multiply by $x^{3/2}$: $x^3 = 8$, so $x = 2$.

Question 8 TRI-02 · Triangles, radians and sectors · spec MM4.1

A triangle has sides 4, 5 and 6. Find the cosine of its largest angle.

Answer: B

The largest angle is opposite the side 6: $\cos \theta = \frac{16 + 25 - 36}{2 \cdot 4 \cdot 5} = \frac{5}{40} = \frac{1}{8}$.

Question 9 PRO-06 · Probability · spec M7.7

In a class of 30, 18 study French, 12 study German and 5 study both. A student is chosen at random. Given that the student studies French, what is the probability that they also study German?

Answer: C

Conditional on French (18 students), 5 also study German: $\frac{5}{18}$.

Question 10 QUA-01 · Quadratic functions · spec MM1.3

The equation $x^2 + (k - 2)x + (k + 1) = 0$ has two distinct real roots. Find the complete set of possible values of k .

Answer: E

Discriminant $(k - 2)^2 - 4(k + 1) = k^2 - 8k = k(k - 8) > 0$, so $k < 0$ or $k > 8$.

Question 11 FUN-02 · Functions and transformations · spec MM8.2

$f(x) = x^2 - 4x$. The graph of $y = f(x + 2)$ has its minimum point at

Answer: A

$f(x) = (x - 2)^2 - 4$ has minimum at $(2, -4)$. Replacing x by $x + 2$ translates the graph 2 units to the left: $(0, -4)$.

Question 12 STA-10 · Statistics · spec M6.3

For which of the following data sets is the mean greater than the median?

Answer: D

A large outlier pulls the mean above the median: mean 6 > median 3. In B, D, E the mean equals the median; in C the mean 9.4 < median 11.

Question 13 DIF-14 · Differentiation · spec MM6.2

Find $\frac{d^2y}{dx^2}$ when $y = \frac{1}{x^2} + 4\sqrt{x}$.

Answer: D

$y' = -2x^{-3} + 2x^{-1/2}$; $y'' = 6x^{-4} - x^{-3/2}$.

Question 14 SEQ-04 · Sequences and series · spec MM2.2

The n th term of an arithmetic sequence is $u_n = 7 - 4n$. Find the sum of the first 30 terms.

Answer: A

$u_1 = 3$, $u_{30} = 7 - 120 = -113$. $S_{30} = \frac{30}{2}(3 - 113) = 15(-110) = -1650$.

Question 15 GRA-08 · Graphs of functions · spec MM8.7

How many real solutions does the equation $x^3 = 3x + 1$ have?

Answer: D

Sketch $y = x^3 - 3x - 1$: turning points at $x = \pm 1$ with $y(-1) = 1 > 0$ and $y(1) = -3 < 0$. A cubic whose local maximum is positive and local minimum is negative crosses the axis three times.

Question 16 SIM-13 · Simultaneous equations and inequalities · spec MM1.5

How many integers x satisfy $x^2 < 5x + 24$ and $x^2 > 3x$?

Answer: E

$x^2 - 5x - 24 < 0 \Rightarrow (x - 8)(x + 3) < 0 \Rightarrow -3 < x < 8$. $x(x - 3) > 0 \Rightarrow x < 0$ or $x > 3$. Integers: $-2, -1$ and $4, 5, 6, 7$: six.

Question 17 GRA-04 · Graphs of functions · spec MM8.7

The line $y = x + c$ and the curve $y = \frac{1}{x}$ have no points of intersection. Find the complete set of values of c .

Answer: F

$x^2 + cx - 1 = 0$ always has discriminant $c^2 + 4 > 0$: two intersections for every c . So there are no values of c with no intersection.

Question 18 SEQ-20 · Sequences and series · spec MM2.2

The integers from 1 to 100 inclusive are written down. What is the sum of those that are multiples of 3 or 5 (or both)?

Answer: E

Multiples of 3: $3 + \dots + 99 = 3(1 + \dots + 33) = 3 \times 561 = 1683$. Multiples of 5: $5(1 + \dots + 20) = 1050$. Multiples of 15: $15(1 + \dots + 6) = 315$. Total $1683 + 1050 - 315 = 2418$.

Question 19 SIM-08 · Simultaneous equations and inequalities · spec MM1.5

Which of the following inequalities has exactly the same solution set as $\frac{1}{x} > 2$?

Answer: B

If $x > 0$: $1 > 2x$, so $0 < x < \frac{1}{2}$. If $x < 0$: $\frac{1}{x}$ is negative and cannot exceed 2. So $0 < x < \frac{1}{2}$. (Multiplying by x without considering its sign is the classic error.)

Question 20 INT-05 · Integration · spec MM7.1

Find the area of the finite region enclosed by the curve $y = x^2$ and the line $y = 2x + 3$.

Answer: E

Intersections: $x^2 - 2x - 3 = (x - 3)(x + 1) = 0$. Area
 $= \int_{-1}^3 (2x + 3 - x^2) dx = \left[x^2 + 3x - \frac{x^3}{3} \right]_{-1}^3 = (9 + 9 - 9) - \left(1 - 3 + \frac{1}{3} \right) = 9 + \frac{5}{3} = \frac{32}{3}$.